

COMP9003 26s1 Final Practice

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W1 Unit Introduction and Programming

Q1. MCQ Semester 1 2021 Final (modified)

Which statement best matches the unit introduction and programming focus?

- (A) A variable, object, or control structure must be traced according to Java evaluation rules.
- (B) The static type decides every runtime method body.
- (C) All Java collections preserve sorted order automatically.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: A Java source file can contain valid helper methods but still have no runnable entry point if no suitable main method is present. Explain.

Q3. MCQ

Which debugging action is most useful before changing code for a unit introduction and programming bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Create a small trace table or failing input that reproduces the issue.
- (C) Delete all branches or loops.
- (D) Change all fields to public.

Q4. Short Answer Warning

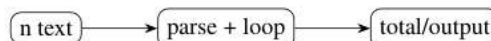
Name two mistakes a beginner might make when writing main or print statements, and give the symptom for each.

Q5. Calculation

State the exact output of this code and explain the key step.

```
int a = 2;
int b = 5;
if (++a * 2 > b) {
    b += a;
} else {
    b -= a;
}
System.out.println(a + ", " + b);
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

A student writes every statement inside main and uses no helper methods. Explain why this can compile but still be poor OO design.

Q8. Calculation Pitfall

For a method that sums integers from 1 to n, give the smallest boundary input worth testing and the expected output.

**Q9. MCQ**

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A paragraph about unrelated Java history.
- (C) A statement of the result plus the Java rule that caused it.
- (D) A claim that all alternatives are possible.

Q10. Long Question Strategy

This multi-part question is about unit introduction and programming.

- Write `public static int sumTo(int n)` returning $1 + \dots + n$, with 0 returned when $n = 0$.
- State the accumulator meaning after each loop iteration.
- Give expected results for $n=0$ and $n=4$.
- Explain the bug if the loop uses $i < n$.



W2 Java Basics and Primitive Data Types

Q1. MCQ Semester 2 2021 Final (modified)

Which statement best matches the java basics and primitive data types focus?

- (A) The static type decides every runtime method body.
- (B) A variable, object, or control structure must be traced according to Java evaluation rules.
- (C) All Java collections preserve sorted order automatically.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: `String r = " + 10 + 20;` stores 30. Explain left-to-right evaluation.

Q3. MCQ

Which debugging action is most useful before changing code for a java basics and primitive data types bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Delete all branches or loops.
- (C) Create a small trace table or failing input that reproduces the issue.
- (D) Change all fields to public.

Q4. Short Answer Warning

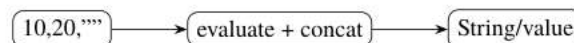
Give two traps involving primitive types, wrappers, integer division, or string concatenation, with a prevention rule for each.

Q5. Calculation

State the exact output of this code and explain the key step.

```
int a = 2;
int b = 5;
if (++a * 2 > b) {
    b += a;
} else {
    b -= a;
}
System.out.println(a + ", " + b);
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Use primitive/reference type confusion to show why compiling Java code is not the same as understanding the program.

Q8. Calculation Pitfall

For `average(int total, int count)`, give a boundary input for count and state the expected behaviour.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A paragraph about unrelated Java history.
- (C) A claim that all alternatives are possible.
- (D) A statement of the result plus the Java rule that caused it.

**Q10. Long Question Strategy**

This multi-part question is about java basics and primitive data types.

- (a) Write static double average(int total,int count) that rejects count 0.
- (b) Explain why total / count may be wrong.
- (c) Give tests for (7,2) and (5,0).
- (d) Say whether double or Double is better here.



W3 Decision and Iteration

Q1. MCQ Semester 1 2022 Final (modified)

Which statement best matches the decision and iteration focus?

- (A) The static type decides every runtime method body.
- (B) All Java collections preserve sorted order automatically.
- (C) A variable, object, or control structure must be traced according to Java evaluation rules.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: Every Java switch case automatically stops after its statements. Justify using fall-through.

Q3. MCQ

Which debugging action is most useful before changing code for a decision and iteration bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Delete all branches or loops.
- (C) Change all fields to public.
- (D) Create a small trace table or failing input that reproduces the issue.

Q4. Short Answer Warning

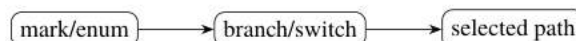
Give two traps involving loop bounds or switch fall-through, with the exact symptom each produces.

Q5. Calculation

State the exact output of this code and explain the key step.

```
int a = 2;
int b = 5;
if (++a * 2 > b) {
    b += a;
} else {
    b -= a;
}
System.out.println(a + ", " + b);
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Explain how branch-heavy code can compile while violating the intended decision table.

Q8. Calculation Pitfall

For a loop over array indexes, give one first-index and one last-index boundary test.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) A statement of the result plus the Java rule that caused it.
- (B) Only the final output, with no reasoning.
- (C) A paragraph about unrelated Java history.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about decision and iteration.

- (a) Complete static String grade(int mark) returning F/P/C/D/HD for standard mark ranges.
- (b) State how invalid marks such as -1 or 101 are handled.
- (c) Give tests for 49, 50, 84, 85.
- (d) Explain why condition ordering matters.



W4 Arrays and Strings

Q1. MCQ Semester 2 2022 Final (modified)

Which statement best matches the arrays and strings focus?

- (A) The static type decides every runtime method body.
- (B) All Java collections preserve sorted order automatically.
- (C) A constructor is a method with an arbitrary return type.
- (D) A variable, object, or control structure must be traced according to Java evaluation rules.

Q2. True/False Pitfall

True or false: If a method assigns an array parameter to a new array, the caller's original array variable now points to the new array.

Q3. MCQ

Which debugging action is most useful before changing code for a arrays and strings bug?

- (A) Create a small trace table or failing input that reproduces the issue.
- (B) Guess a fix and rewrite the whole class.
- (C) Delete all branches or loops.
- (D) Change all fields to public.

Q4. Short Answer Warning

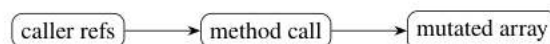
Give two traps involving array indexing, array length, or reference aliasing.

Q5. Calculation

State the exact output of this code and explain the key step.

```
int[] left = {3, 6, 9};
int[] right = {1, 2, 3};
change(left, right);
System.out.println(left[0] + "," + left[1] + "," + left[2]);
System.out.println(right[0] + "," + right[1] + "," + right[2]);
static void change(int[] a, int[] b) {
    b[1] = a[0];
    a = b;
    a[2] = 99;
}
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Use array aliasing to show how code can compile but still be logically wrong.

Q8. Calculation Pitfall

For `first(int[] values)`, give the boundary input that should be rejected and the expected exception or precondition.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A statement of the result plus the Java rule that caused it.
- (C) A paragraph about unrelated Java history.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about arrays and strings.

- (a) Write static int countEven(int[] values).
- (b) State or handle the null-input precondition.
- (c) Trace the method for {2,3,4,7}.
- (d) Explain why the loop condition must be $i < \text{values.length}$.



W5 Introduction to OOP

Q1. MCQ Semester 1 2023 Final (modified)

Which statement best matches the introduction to oop focus?

- (A) A variable, object, or control structure must be traced according to Java evaluation rules.
- (B) The static type decides every runtime method body.
- (C) All Java collections preserve sorted order automatically.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: A class with all public fields is automatically well-encapsulated because clients can access the data directly.

Q3. MCQ

Which debugging action is most useful before changing code for a introduction to oop bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Create a small trace table or failing input that reproduces the issue.
- (C) Delete all branches or loops.
- (D) Change all fields to public.

Q4. Short Answer Warning

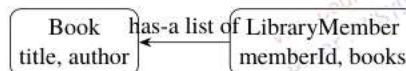
Give two traps involving constructors, this, or field visibility in a simple class.

Q5. Calculation

State the exact output of this code and explain the key step.

```
ArrayList<Integer> nums = new ArrayList<>();
for (int i = 0; i < 5; i++) nums.add(i * 3);
nums.remove(2);
nums.add(1, 20);
System.out.println(nums);
```

Q6. Table/Diagram Novel



Using the diagram, explain why LibraryMember should contain an ArrayList<Book> rather than extend Book. Then name one invariant for each class.

Q7. Short Answer

Use a Book class with public mutable fields as an example of code that compiles but violates encapsulation.

Q8. Calculation Pitfall

For Book(String title, String author), give a boundary input and expected validation behaviour.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A paragraph about unrelated Java history.
- (C) A statement of the result plus the Java rule that caused it.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about introduction to oop.

- (a) Design Book with private title and author, a constructor, and toString.
- (b) State one invariant for Book.
- (c) Write a two-line tester that constructs and prints a book.
- (d) Explain why toString returns a string rather than printing.



W6 Fields Constructors and Methods

Q1. MCQ Semester 2 2023 Final (modified)

Which statement best matches the fields constructors and methods focus?

- (A) The static type decides every runtime method body.
- (B) A variable, object, or control structure must be traced according to Java evaluation rules.
- (C) All Java collections preserve sorted order automatically.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: A static method can directly use `this` if it is called from inside the same class.

Q3. MCQ

Which debugging action is most useful before changing code for a fields constructors and methods bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Delete all branches or loops.
- (C) Create a small trace table or failing input that reproduces the issue.
- (D) Change all fields to public.

Q4. Short Answer Warning

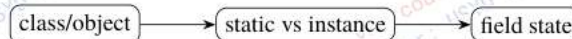
Give two traps involving shadowed fields, static methods, or overloaded method selection.

Q5. Calculation

State the exact output of this code and explain the key step.

```
ArrayList<Integer> nums = new ArrayList<>();
for (int i = 0; i < 5; i++) nums.add(i * 3);
nums.remove(2);
nums.add(1, 20);
System.out.println(nums);
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Explain how a class with only static methods can compile while failing to model per-object state.

Q8. Calculation Pitfall

For overloaded methods `add(int)` and `add(String)`, give one call that selects each overload.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A paragraph about unrelated Java history.
- (C) A claim that all alternatives are possible.
- (D) A statement of the result plus the Java rule that caused it.

**Q10. Long Question Strategy**

This multi-part question is about fields constructors and methods.

- (a) Create a Counter class with private count, instance increment(), and static created.
- (b) Describe what the constructor updates.
- (c) State the state after constructing two counters and incrementing one.
- (d) Explain why increment should not be static.



W7 Inheritance and Encapsulation

Q1. MCQ Semester 1 2024 Final (modified)

Which statement best matches the inheritance and encapsulation focus?

- (A) The static type decides every runtime method body.
- (B) All Java collections preserve sorted order automatically.
- (C) A variable, object, or control structure must be traced according to Java evaluation rules.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: A subclass constructor body executes before the superclass constructor.

Q3. MCQ

Which debugging action is most useful before changing code for a inheritance and encapsulation bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Delete all branches or loops.
- (C) Change all fields to public.
- (D) Create a small trace table or failing input that reproduces the issue.

Q4. Short Answer Warning

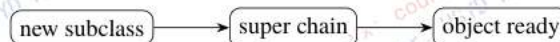
Give two traps involving extends, super(...), access modifiers, or overriding.

Q5. Calculation

State the exact output of this code and explain the key step.

```
class Animal { String sound() { return "A"; } }
class Cat extends Animal { String sound() { return "C" + super.sound(); } }
class Kitten extends Cat { String sound() { return super.sound() + "K"; } }
Animal a = new Kitten();
System.out.println(a.sound());
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Give an example where inheritance compiles but is poor design because the relationship is not truly is-a.

Q8. Calculation Pitfall

For a superclass without a no-argument constructor, state the constructor error a subclass may hit.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) A statement of the result plus the Java rule that caused it.
- (B) Only the final output, with no reasoning.
- (C) A paragraph about unrelated Java history.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about inheritance and encapsulation.

- (a) Design `Person` and `Student` extends `Person` with a name initialised through constructors.
- (b) Write the needed `super(...)` call.
- (c) Explain constructor execution order.
- (d) State why direct access to a private parent field fails.



W8 Polymorphism Abstract Classes Interfaces

Q1. MCQ Semester 2 2024 Final (modified)

Which statement best matches the polymorphism abstract classes interfaces focus?

- (A) The static type decides every runtime method body.
- (B) All Java collections preserve sorted order automatically.
- (C) A constructor is a method with an arbitrary return type.
- (D) A variable, object, or control structure must be traced according to Java evaluation rules.

Q2. True/False Pitfall

True or false: If `Shape s = new Triangle();`, then `s` can call every public method declared only in `Triangle`.

Q3. MCQ

Which debugging action is most useful before changing code for a polymorphism abstract classes interfaces bug?

- (A) Create a small trace table or failing input that reproduces the issue.
- (B) Guess a fix and rewrite the whole class.
- (C) Delete all branches or loops.
- (D) Change all fields to public.

Q4. Short Answer Warning

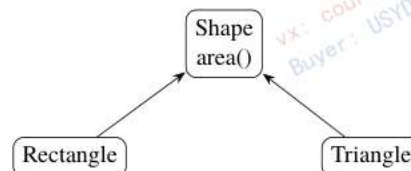
Give two traps involving static type, dynamic type, abstract classes, interfaces, or casting.

Q5. Calculation

State the exact output of this code and explain the key step.

```
class Animal { String sound() { return "A"; } }
class Cat extends Animal { String sound() { return "C" + super.sound(); } }
class Kitten extends Cat { String sound() { return super.sound() + "K"; } }
Animal a = new Kitten();
System.out.println(a.sound());
```

Q6. Table/Diagram Novel



For `Shape s = new Triangle();`, identify which methods may be called through `s`, and which implementation of `area()` runs.

Q7. Short Answer

Explain why replacing polymorphism with many instanceof checks may compile but be weaker design.

Q8. Calculation Pitfall

For a method accepting `Shape`, give one test with a `Rectangle` and one with a `Triangle`.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A statement of the result plus the Java rule that caused it.
- (C) A paragraph about unrelated Java history.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about polymorphism abstract classes interfaces.

- (a) Define interface Shape with double area().
- (b) Sketch Rectangle implements Shape with width and height.
- (c) For Shape s = new Rectangle(2,3), state allowed calls and area result.
- (d) Explain when a cast to Rectangle is unsafe.



W9 Class Members and ArrayList

Q1. MCQ Semester 1 2021 Final (modified)

Which statement best matches the class members and arraylist focus?

- (A) A variable, object, or control structure must be traced according to Java evaluation rules.
- (B) The static type decides every runtime method body.
- (C) All Java collections preserve sorted order automatically.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: HashMap keys are always printed in sorted order.

Q3. MCQ

Which debugging action is most useful before changing code for a class members and arraylist bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Create a small trace table or failing input that reproduces the issue.
- (C) Delete all branches or loops.
- (D) Change all fields to public.

Q4. Short Answer Warning

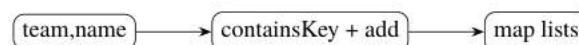
Give two traps involving ArrayList.remove, generic type arguments, or missing HashMap keys.

Q5. Calculation

State the exact output of this code and explain the key step.

```
int[] left = {3, 6, 9};
int[] right = {1, 2, 3};
change(left, right);
System.out.println(left[0] + "," + left[1] + "," + left[2]);
System.out.println(right[0] + "," + right[1] + "," + right[2]);
static void change(int[] a, int[] b) {
    b[1] = a[0];
    a = b;
    a[2] = 99;
}
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Show how using a raw ArrayList can compile with warnings but fail later by storing the wrong type.

Q8. Calculation Pitfall

For grouping names by team number, give the boundary case of the first name for a new team.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A paragraph about unrelated Java history.
- (C) A statement of the result plus the Java rule that caused it.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about class members and arraylist.

- (a) Write the core update for `HashMap<Integer, ArrayList<String>>` teams adding one team/name pair.
- (b) Explain why unseen teams need a new list.
- (c) State the map after adding (2,Ana), (2,Bo), (3,Cid).
- (d) Explain why `HashMap<int, String>` is invalid.



W10 OO Software Development and Testing

Q1. MCQ Semester 2 2022 Final (modified)

Which statement best matches the oo software development and testing focus?

- (A) The static type decides every runtime method body.
- (B) A variable, object, or control structure must be traced according to Java evaluation rules.
- (C) All Java collections preserve sorted order automatically.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: A black-box test must inspect the private fields of the class under test.

Q3. MCQ

Which debugging action is most useful before changing code for a oo software development and testing bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Delete all branches or loops.
- (C) Create a small trace table or failing input that reproduces the issue.
- (D) Change all fields to public.

Q4. Short Answer Warning

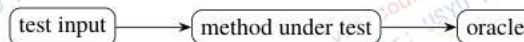
Give two traps involving weak tests, missing boundary cases, or broken class invariants.

Q5. Calculation

State the exact output of this code and explain the key step.

```
ArrayList<Integer> nums = new ArrayList<>();
for (int i = 0; i < 5; i++) nums.add(i * 3);
nums.remove(2);
nums.add(1, 20);
System.out.println(nums);
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Explain why passing one normal test does not prove an OO method is correct.

Q8. Calculation Pitfall

For a method that removes an item by index, give tests for index 0, last valid index, and invalid index.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A paragraph about unrelated Java history.
- (C) A claim that all alternatives are possible.
- (D) A statement of the result plus the Java rule that caused it.

**Q10. Long Question Strategy**

This multi-part question is about oo software development and testing.

- (a) For boolean `hasShortName(ArrayList<String> names)`, write three black-box tests.
- (b) Include a normal false case, a true case, and a null case if nulls are ignored.
- (c) State each expected result.
- (d) Explain which branch or boundary each test covers.



W11 Recursion Exceptions and Debugging

Q1. MCQ Semester 1 2024 Final (modified)

Which statement best matches the recursion exceptions and debugging focus?

- (A) The static type decides every runtime method body.
- (B) All Java collections preserve sorted order automatically.
- (C) A variable, object, or control structure must be traced according to Java evaluation rules.
- (D) A constructor is a method with an arbitrary return type.

Q2. True/False Pitfall

True or false: A recursive method with a base case can never overflow the stack.

Q3. MCQ

Which debugging action is most useful before changing code for a recursion exceptions and debugging bug?

- (A) Guess a fix and rewrite the whole class.
- (B) Delete all branches or loops.
- (C) Change all fields to public.
- (D) Create a small trace table or failing input that reproduces the issue.

Q4. Short Answer Warning

Give two traps involving missing base cases, non-progressing recursion, or catch block ordering.

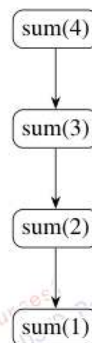
Q5. Calculation

State the exact output of this code and explain the key step.

```
static int f(int n) {
    if (n <= 1) return 1;
    return n + f(n / 2);
}
```

```
System.out.println(f(10));
```

Q6. Table/Diagram Novel



Annotate the base case and the return values on the call-stack diagram for a recursive method that returns $1 + \dots + n$.

Q7. Short Answer

Explain how exception handling can compile while hiding a bug if it catches `Exception` too broadly.

**Q8. Calculation Pitfall**

For recursive gcd, give the boundary case that stops recursion.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) A statement of the result plus the Java rule that caused it.
- (B) Only the final output, with no reasoning.
- (C) A paragraph about unrelated Java history.
- (D) A claim that all alternatives are possible.

Q10. Long Question Strategy

This multi-part question is about recursion exceptions and debugging.

- Write recursive `gcd(int a, int b)` using Euclid's rule.
- Use it to compute `lcm(int a, int b)`.
- Trace `gcd(12, 8)`.
- State one precondition involving zero or negative inputs.



W12 Revision and Exam Integration

Q1. MCQ Semester 2 2024 Final (modified)

Which statement best matches the revision and exam integration focus?

- (A) The static type decides every runtime method body.
- (B) All Java collections preserve sorted order automatically.
- (C) A constructor is a method with an arbitrary return type.
- (D) A variable, object, or control structure must be traced according to Java evaluation rules.

Q2. True/False Pitfall

True or false: In an enum that implements an interface, the implemented method may be private if it is only used inside the enum.

Q3. MCQ

Which debugging action is most useful before changing code for a revision and exam integration bug?

- (A) Create a small trace table or failing input that reproduces the issue.
- (B) Guess a fix and rewrite the whole class.
- (C) Delete all branches or loops.
- (D) Change all fields to public.

Q4. Short Answer Warning

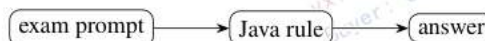
Give two final-exam traps that combine enum/interface syntax, generics, collections, or dispatch.

Q5. Calculation

State the exact output of this code and explain the key step.

```
ArrayList<Integer> nums = new ArrayList<>();
for (int i = 0; i < 5; i++) nums.add(i * 3);
nums.remove(2);
nums.add(1, 20);
System.out.println(nums);
```

Q6. Table/Diagram Novel



Use the diagram to explain the key state change or Java rule for this week's topic.

Q7. Short Answer

Explain why an answer that only says 'polymorphism' without naming static and dynamic type may lose marks.

Q8. Calculation Pitfall

For a design using a list of enum values, give one empty-list test and one multi-element test.

Q9. MCQ

Which answer best describes a high-quality final-exam explanation?

- (A) Only the final output, with no reasoning.
- (B) A statement of the result plus the Java rule that caused it.
- (C) A paragraph about unrelated Java history.
- (D) A claim that all alternatives are possible.

**Q10. Long Question Strategy**

This multi-part question is about revision and exam integration.

- (a) Correct an enum `Assessment` implements `Weighted` with `QUIZ`, `FINAL`, and `weight()`.
- (b) State why `extends Weighted` is wrong.
- (c) Create an `ArrayList<Assessment>` and add both constants.
- (d) Explain why the interface method must be public.



Answer Key

W1 Unit Introduction and Programming

- Q1. A. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. B. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. The output is 3,8. `++a` changes `a` to 3 before multiplication; `3*2>5` is true, so `b` becomes 8.
- Q6. Example labels: input is `n` or command-line text; process is `parse/validate/sum`; state is variables such as `i` and `total`. A useful test checks the exact printed output and `n=0`.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. C. For marks, state the result and the rule: `dispatch`, `aliasing`, `loop bounds`, `fall-through`, `access`, or `exception handling`.
- Q10. Use `total=0` and loop `i` from 1 through `n` inclusive. The accumulator is the sum of processed integers. `n=0` returns 0; `n=4` returns 10. Using `i<n` omits `n`, so `n=4` gives 6.

W2 Java Basics and Primitive Data Types

- Q1. B. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. C. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. The output is 3,8. `++a` changes `a` to 3 before multiplication; `3*2>5` is true, so `b` becomes 8.
- Q6. Labels should show input literals/variables, the conversion or concatenation process, and the final value plus type.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. D. For marks, state the result and the rule: `dispatch`, `aliasing`, `loop bounds`, `fall-through`, `access`, or `exception handling`.
- Q10. If `count==0` throw `IllegalArgumentException`; otherwise return `(double) total / count`. (7,2) gives 3.5; (5,0) throws. Primitive double is enough because no null result is needed.

W3 Decision and Iteration

- Q1. C. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. D. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. The output is 3,8. `++a` changes `a` to 3 before multiplication; `3*2>5` is true, so `b` becomes 8.
- Q6. Labels should show input condition, branch/loop process, and resulting output or variable state. Include boundary branch tests.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. A. For marks, state the result and the rule: `dispatch`, `aliasing`, `loop bounds`, `fall-through`, `access`, or `exception handling`.
- Q10. Validate 0..100 first, then use ordered if/else ranges: 49->F, 50->P, 84->D, 85->HD. Ordering matters so broad ranges do not capture later cases.



W4 Arrays and Strings

- Q1. D. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. A. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. left prints 3,6,9 and right prints 1,3,99. The assignment `a=b` is local, then `a[2]` mutates the array also referenced by right.
- Q6. Show caller and parameter references pointing at the same array. Element writes persist; parameter reassignment changes only the local arrow.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. B. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. Loop from `i=0` to `values.length-1` and count `values[i]%2==0`. State `values!=null` or explicitly reject null. 2,3,4,7 gives 2. Using `<= length` accesses an invalid index.

W5 Introduction to OOP

- Q1. A. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. B. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. [0, 20, 3, 9, 12]. `remove(2)` removes the value at index 2, then `add(1,20)` inserts before current index 1.
- Q6. `LibraryMember` has-a list of books; it is not a kind of book, so composition is correct. Invariants: `Book` has non-null title/author; `LibraryMember` has non-null `memberId/name` and non-null `borrowedBooks`.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. C. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. Fields are private strings set by the constructor. Invariants: title and author non-null/non-empty. `toString` returns e.g. `title + " by " + author`, making it usable by print, concatenation, and tests.

W6 Fields Constructors and Methods

- Q1. B. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. C. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. [0, 20, 3, 9, 12]. `remove(2)` removes the value at index 2, then `add(1,20)` inserts before current index 1.
- Q6. Separate class-level/static state from object-level state and show which operation updates each.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. D. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. `count` belongs to each object; `created` belongs to the class and increments in the constructor. After two constructions `created=2`; only one object's count changes. `increment` needs `this.count`, so it is an instance method.



W7 Inheritance and Encapsulation

- Q1. C. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. D. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. CAK. Dynamic dispatch starts at Kitten.sound(), then super calls Cat.sound(), which calls Animal.sound().
- Q6. The arrow should represent a genuine is-a relationship. Constructor and method reasoning still follow Java access and dispatch rules.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. A. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. Student constructor calls super(name) first. Parent construction runs before student-specific initialisation. Private parent fields are not directly accessible in subclasses; use parent methods/accessors.

W8 Polymorphism Abstract Classes Interfaces

- Q1. D. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. A. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. CAK. Dynamic dispatch starts at Kitten.sound(), then super calls Cat.sound(), which calls Animal.sound().
- Q6. Through Shape s, only Shape-declared methods are callable. The dynamic object is Triangle, so Triangle's area() implementation runs.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. B. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. Shape declares area; Rectangle stores dimensions and returns width*height. Through Shape, only Shape-declared calls are allowed; Rectangle.area runs and returns 6.0. Casting is unsafe unless the runtime object really is Rectangle.

W9 Class Members and ArrayList

- Q1. A. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. B. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. left prints 3,6,9 and right prints 1,3,99. The assignment a=b is local, then a[2] mutates the array also referenced by right.
- Q6. Show key input, grouping process, and map state after insertion. Tests should include a new key and an existing key.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. C. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. If the key is absent, put a new ArrayList, then teams.get(team).add(name). Result: 2->[Ana, Bo], 3->[Cid]. Generic type arguments must be reference types, so use Integer.



W10 OO Software Development and Testing

- Q1. B. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. C. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. [0, 20, 3, 9, 12]. remove(2) removes the value at index 2, then add(1,20) inserts before current index 1.
- Q6. Connect input cases to process and expected result/state. A strong test has a clear oracle and covers a distinct branch or boundary.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. D. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. Example tests: [Al, Mira]->true; [Sam, Noah]->false; [null, Bo]->true if nulls are ignored. They cover true detection, no match, and null-handling plus later match.

W11 Recursion Exceptions and Debugging

- Q1. C. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. D. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. 18. The calls use $n=10, 5, 2, 1$, so the result is $10+5+2+1$.
- Q6. Base case is sum(1) or sum(0), depending on definition. For $1+\dots+n$, returns unwind as sum(1)=1, sum(2)=3, sum(3)=6, sum(4)=10.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. A. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. gcd(a,b) returns a when $b=0$, otherwise gcd(b,a%b). gcd(12,8)->gcd(8,4)->gcd(4,0)=4. lcm can be $a/\text{gcd}(a,b)*b$. State positive-input precondition or define zero/negative handling.

W12 Revision and Exam Integration

- Q1. D. The question is modified from the listed final source and targets the same kind of Java reasoning rather than the original wording.
- Q2. Award full credit for the correct true/false judgement plus the relevant Java rule for this week.
- Q3. A. A minimal reproducible trace separates the actual fault from surrounding noise.
- Q4. Full-credit answers name two concrete traps from the week and give a prevention rule or symptom for each.
- Q5. [0, 20, 3, 9, 12]. remove(2) removes the value at index 2, then add(1,20) inserts before current index 1.
- Q6. Tie input/topic, Java rule, and state/output. A complete integrated answer names the specific rule, not only the broad topic.
- Q7. Compiling is only syntactic/type correctness. A good answer gives the requested topic-specific example and explains the design, runtime, or semantic failure.
- Q8. The answer must give a concrete input/object state and a precise expected result, state, or exception, with a brief boundary justification.
- Q9. B. For marks, state the result and the rule: dispatch, aliasing, loop bounds, fall-through, access, or exception handling.
- Q10. Use enum Assessment implements Weighted with constants and a private weight field; public double weight() returns it. Enums implement interfaces, not extend them. Use new ArrayList<>() and add constants. Interface methods are public by contract.